

Industrial Oriented Mini Project Report (CM653PC)

on

“Bhagavad Gita Guidance System using Constraint-Aware Retrieval”

Submitted

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in

Computer Science and Engineering (AI & ML)

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CERTIFICATE



This is to certify that the Industrial Oriented Mini Project (CM653PC) entitled "**Bhagavad Gita Guidance System using Constraint-Aware Retrieval**" is being submitted by **PARSI SRIRAM (23261A6639)** and **VEERA NAGA SAI PRANAV SIDDHARTHA TAMADA(23261A6660)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in Computer Science Engineering – Artificial Intelligence and Machine Learning from Mahatma Gandhi Institute of Technology (A), Hyderabad, during the academic year 2025–2026.

The results embodied in this project have not been submitted to any other University or Institute for the award of any degree or diploma.

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DECLARATION

This is to certify that the work reported in this project titled "**Bhagavad Gita Guidance System using Constraint-Aware Retrieval**" is a record of work done by us in the Department of Computer Science and Engineering (Artificial intelligence and Machine Learning), Mahatma Gandhi Institute of Technology, Hyderabad.

No part of the work is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred in the text. The report is based on the work done entirely by us and not copied from any other source.

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ABSTRACT

In the era of artificial intelligence, the need for personalized, context-aware guidance systems is rapidly increasing, especially in domains related to mental well-being and decision-making. While many individuals turn to spiritual texts such as the Bhagavad Gita for clarity and emotional support, identifying relevant verses and interpreting them in a practical, actionable manner remains a significant challenge. Existing AI systems based on Retrieval-Augmented Generation (RAG) primarily rely on semantic similarity and often overlook critical user-specific constraints such as emotional context, situational intent, and desired form of guidance. As a result, the outputs are frequently theoretical, generic, and lack real-world applicability. This project presents a constraint-aware, practice-oriented Bhagavad Gita AI system that enables users to receive meaningful, context-sensitive, and actionable guidance derived from classical philosophical knowledge.

The proposed methodology integrates hybrid retrieval techniques combining metadata-driven filtering (emotion, life situation, and principles) with semantic matching, along with a structured reasoning pipeline that maps user queries through stages of emotion identification, situational understanding, and principle extraction before selecting relevant verses. A robust fallback reasoning mechanism ensures reliable outputs even in the absence of large language model responses, enabling consistent offline functionality. Unlike conventional systems, the model extends beyond information retrieval by generating concise insights, personalized advice, and micro-practices that users can apply in real life. The system is implemented using a Node.js-based pipeline with optional local LLM integration via Ollama (Mistral) and supports further enhancement through Python-based semantic retrieval using FAISS and embeddings. The results demonstrate that the system provides accurate, diverse, and actionable guidance while maintaining high reliability and responsiveness, thereby bridging the gap between philosophical knowledge and practical application.

Keywords: *Bhagavad Gita, Hybrid RAG, Practice-Oriented Systems, Offline AI, Semantic Search, Knowledge-Based Guidance, Human-Centered AI*

1. INTRODUCTION

Modern digital systems provide access to vast philosophical knowledge, yet users struggle to derive relevant, personalized guidance. Although the Bhagavad Gita offers deep insights into emotions, decision-making, and life challenges, navigating its verses requires prior knowledge of chapters and context. With advancements in Large Language Models (LLMs) and intelligent retrieval systems, it is now possible to build AI solutions that understand natural language queries, interpret emotional context, and deliver meaningful insights.

This project presents a Constraint-Aware, Practice-Oriented Bhagavad Gita AI System, an offline-capable assistant that analyzes user queries, identifies emotional and situational context, and retrieves relevant verses using hybrid retrieval techniques. It combines structured reasoning, knowledge mapping, and fallback mechanisms to ensure reliable outputs, even without cloud dependency. Beyond providing insights, the system generates actionable micro-practices, enabling users to apply philosophical guidance in real-life situations.

1.1 Motivation

Three factors motivate the development of this system

1. Accessibility of Knowledge

Users often struggle to locate relevant verses without prior familiarity with the Bhagavad Gita's structure.

2. Practical Application Gap

Existing systems provide theoretical explanations but lack actionable guidance for real-life situations.

3. Personalized Emotional Support

Users require context-aware responses based on their emotional state (anxiety, confusion, anger, etc.), which traditional systems fail to address effectively.

1.2 Problem Definition

Current systems for spiritual guidance struggle to map user queries to relevant Bhagavad Gita verses without explicit references, lack awareness of emotional and situational context, and rely heavily on semantic similarity, resulting in outputs that are informative but not actionable. Therefore, there is a need for an intelligent system that can understand natural language queries with emotional context, map user intent to philosophical principles, and deliver reliable, actionable guidance while operating efficiently in offline or low-resource environments.

1.3 Existing System

1. Manual Study of Bhagavad Gita

Requires prior knowledge of chapters and slokas; time-consuming and non-interactive

2. Basic Search-Based Applications

Keyword-based retrieval with limited contextual understanding

3. RAG-Based AI Systems

Use semantic similarity but often ignore emotional constraints and user intent

4. General AI Chatbots (ChatGPT, Gemini)

Provide explanations but lack structured retrieval and consistent grounding in authentic sources

1.4 Proposed System

The proposed system is a **hybrid, constraint-aware AI pipeline** that integrates retrieval, reasoning, and action generation.

Key Objectives:

- **Natural Language Understanding**

Interpret user queries to extract emotion and situation

- **Hybrid Retrieval System**

Combine metadata (emotion, life situation, principles) with semantic similarity

- **Structured Reasoning Pipeline**

Map:

Query → *Emotion* → *Situation* → *Principle* → *Verse* → *Insight*

- **Fallback Intelligence**

Ensure reliable outputs even when LLM fails

- **Practice-Oriented Output**

Generate actionable micro-practices (5–10 minutes) for real-life application

- **Offline Capability**

Use local models (Mistral via Ollama) with minimal dependency on cloud APIs

1.5 Requirements Specification

Software Requirements

Language: JavaScript (Node.js), optional Python for enhancements

Runtime Environment: Node.js (v18+)

IDE: Visual Studio Code

Local LLM Runtime: Ollama (tinyllama)

Core Modules:

File handling (fs)

HTTP requests (fetch/axios)

Express.js (for API)

Optional Python Stack (for enhancement):

FastAPI

SentenceTransformers

FAISS

Hardware Requirements

- RAM: 8 GB minimum, 16 GB recommended
- Storage: 10 GB free (Tinyllama $\approx$ 4.4 GB weight)
- Processor: Intel Core i5 (8th gen) / AMD Ryzen 5 or higher
- GPU (optional): For faster local inference

2. LITERATURE SURVEY

Research in emotion-aware conversational AI and retrieval-augmented systems has significantly evolved between 2020 and 2026. The literature surveyed spans three key directions: (a) emotion-aware mental health chatbots, (b) retrieval-augmented and LLM-based conversational systems, and (c) evaluation and real-world deployment of AI support systems.

- S. Dessai et al. – Adaptive Emotion Aware Chatbot for Mental Health Using Dynamic Questionnaires (2026) — proposes a chatbot integrating standardized psychological assessments such as PHQ-9 and GAD-7 for adaptive interaction; highlights challenges in user response reliability [1].
- Revisiting AI Based Dialogue Systems for Emotional Support (2026) — provides a comprehensive review of LLM-based emotional support systems; identifies issues related to safety, hallucination, and lack of reliability [2].
- Evans Ankomaah et al. – Emotion Aware AI Chatbots for Mental Health Support in Low Resource Public Health Systems A Case Study from Ghana (2025) — explores deployment of emotion-aware chatbots in low-resource environments; limited scalability across diverse populations [3].
- P. Bagane et al. – AI Powered Mental Health Application with Sentiment Aware NLP and Mood Tracking (2025) — integrates sentiment analysis with mood tracking for continuous monitoring; raises concerns regarding data privacy and security [4].
- X. Feng et al. – Effectiveness of AI Chatbots in Alleviating Mental Health Symptoms A Systematic Review and Meta Analysis (2025) — evaluates clinical effectiveness of chatbot-based interventions; finds inconsistent outcomes across models [5].
- Anandi Dutta et al. – Mentalic Net Development of RAG Based Conversational AI for Mental Health Support (2025) — introduces a retrieval-augmented conversational system for mental health guidance; requires human supervision for safe deployment [6].
- Trishala Ahalpara – Tell Me LLM Powered Mental Well Being Assistant with Retrieval Augmented Generation and Agentic Planning (2025) — combines RAG with planning-based agents for personalized responses; cannot fully replace professional therapy [7].

- Young Min Cho et al. – An Integrative Survey on Mental Health Conversational Agents to Bridge Computer Science and Medical Perspectives (2023) — surveys interdisciplinary approaches to mental health chatbots; lacks implementation-level contributions [8].

- Sarada Devaram – Empathic Chatbot Emotional Intelligence for Mental Health Well being (2020) — focuses on incorporating emotional intelligence in chatbot responses; limited scalability and real-world validation [9].

- Patrick Lewis et al. – Retrieval Augmented Generation for Knowledge Intensive Natural Language Processing Tasks (2020) — introduces the RAG framework combining retrieval with generation; lacks emotional awareness and domain-specific adaptation [10].

Table 2.1 Literature Survey on Guidance System using Constraint-Aware Retrieval

S.NO	AUTHOR / YEAR	TITLE	FOCUS	LIMITATIONS
1	S. Dessai et al., 2026	Adaptive Emotion Aware Chatbot for Mental Health Using Dynamic Questionnaires	Uses adaptive questionnaires like PHQ-9 and GAD-7 for emotion-aware interaction	Accuracy depends on user input reliability
2	2026 Review Study	Revisiting AI Based Dialogue Systems for Emotional Support	Reviews modern LLM-based emotional support systems	Highlights lack of reliability and safety
3	Evans Ankomaah et al., 2025	Emotion Aware AI Chatbots for Mental Health Support in Low Resource Public Health Systems A Case Study from Ghana	Emotion detection with AI chatbot support	Limited generalization across cultures
4	P. Bagane et al., 2025	AI Powered Mental Health Application with Sentiment Aware NLP and Mood Tracking	Combines mood tracking and sentiment-aware NLP	Privacy and data security concerns

S.NO	AUTHOR / YEAR	TITLE	FOCUS	LIMITATIONS
5	X. Feng et al., 2025	Effectiveness of AI Chatbots in Alleviating Mental Health Symptoms A Systematic Review and Meta Analysis	Evaluates effectiveness of AI chatbots in mental health	Inconsistent performance of generative models
6	Anandi Dutta et al., 2025	Mentalic Net Development of RAG Based Conversational AI for Mental Health Support	RAG-based chatbot for mental health support	Requires human oversight and safety validation
7	Trishala Ahalpara, 2025	Tell Me LLM Powered Mental Well Being Assistant with Retrieval Augmented Generation and Agentic Planning	Combines RAG with planning for personalized guidance	Cannot replace professional therapy
8	Young Min Cho et al., 2023	An Integrative Survey on Mental Health Conversational Agents to Bridge Computer Science and Medical Perspectives	Survey of conversational agents in mental health	Lacks implementation-level detail
9	Sarada Devaram, 2020	Empathic Chatbot Emotional Intelligence for Mental Health Well being	Focuses on emotional intelligence in chatbots	Limited scalability and real-world validation
10	Patrick Lewis et al., 2020	Retrieval Augmented Generation for Knowledge Intensive Natural Language Processing Tasks	Introduces RAG framework combining retrieval and generation	Not emotion-aware and lacks domain-specific adaptation

3. DESIGN AND METHODOLOGY

3.1 System Architecture

The architecture is designed to overcome key limitations of traditional keyword-based or purely generative systems by tightly integrating semantic retrieval with context-aware response generation. The Processing Layer ensures that emotionally nuanced or ambiguous queries are interpreted correctly, while the Retrieval Layer grounds responses in authentic scriptural knowledge, reducing hallucinations commonly associated with large language models. The Generation Layer further refines this information into user-friendly explanations, often aligning philosophical teachings with practical, real-world situations. This layered design also enables efficient offline or low-resource deployment by optimizing embedding storage and retrieval mechanisms, making the system scalable, reliable, and suitable for personalized spiritual assistance.

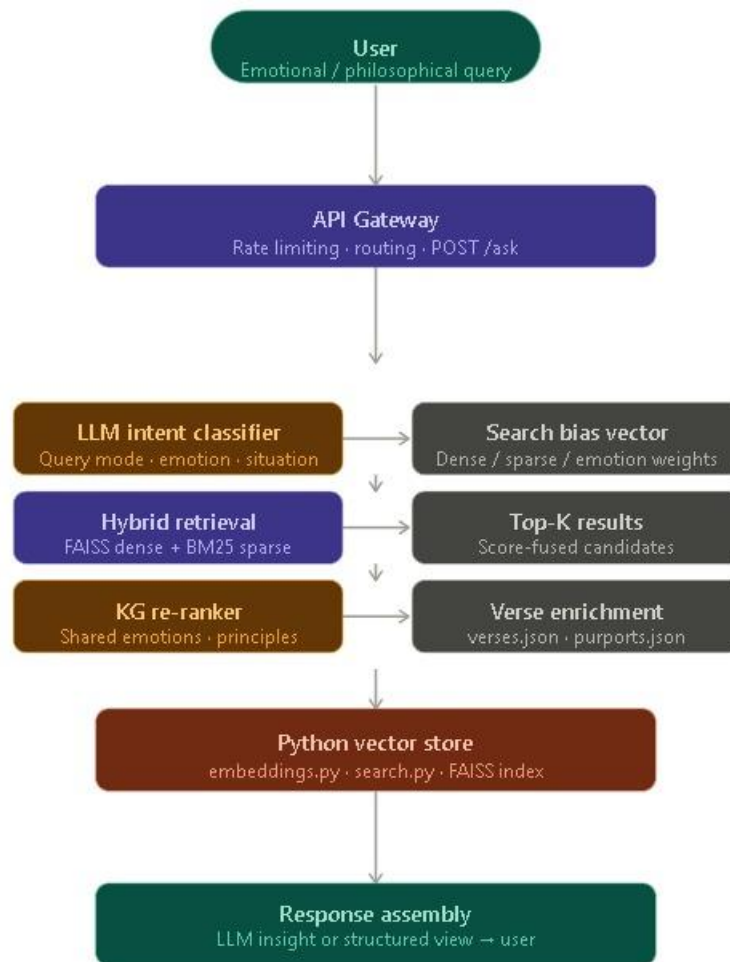


Figure 3.1 System Architecture

The system is organised as a four-layer pipeline:

- (1) **User Interface Layer** that accepts natural language queries and displays contextual spiritual guidance
- (2) **Processing Layer** that performs query understanding, emotion detection, and intent extraction;
- (3) **Retrieval Layer** that maps the processed query to relevant verses using embeddings and semantic search
- (4) **Generation Layer** that produces meaningful, actionable responses using a hybrid Retrieval-Augmented Generation (RAG) approach

3.2 Tools and Technologies

- **Node.js & Express.js** primary backend server handles API routing, rate-limiting, and orchestrating the AI pipeline.
- **Python 3** backend service dedicated to dense vector operations and embedding generation.
- **Ollama local LLM** serving; default model llama3.2:1b (configured via env variables) used for intent classification and text generation.
- **FAISS (faiss-cpu & faiss-node) vector database** powers the dense retrieval of Bhagavad Gita verses based on semantic similarity.
- **FastAPI & Uvicorn Python** web framework serves the isolated REST endpoints for vector search and embeddings.
- **Sentence-Transformers** Python library used to compute text embeddings for the verses and user queries.
- **Vanilla HTML/CSS/JS frontend** lightweight, zero-dependency UI using CSS variables for light/dark theming and native DOM manipulation.
- **node-fetch** server-side HTTP client handles the requests to the Ollama server and the Python FastAPI backend.

4. RESULTS

The main interface of the Bhagavad Gita AI guide where the user inputs a query related to jealousy. The interface includes navigation options and a call-to-action button to request guidance based on the user's emotional or philosophical concern.

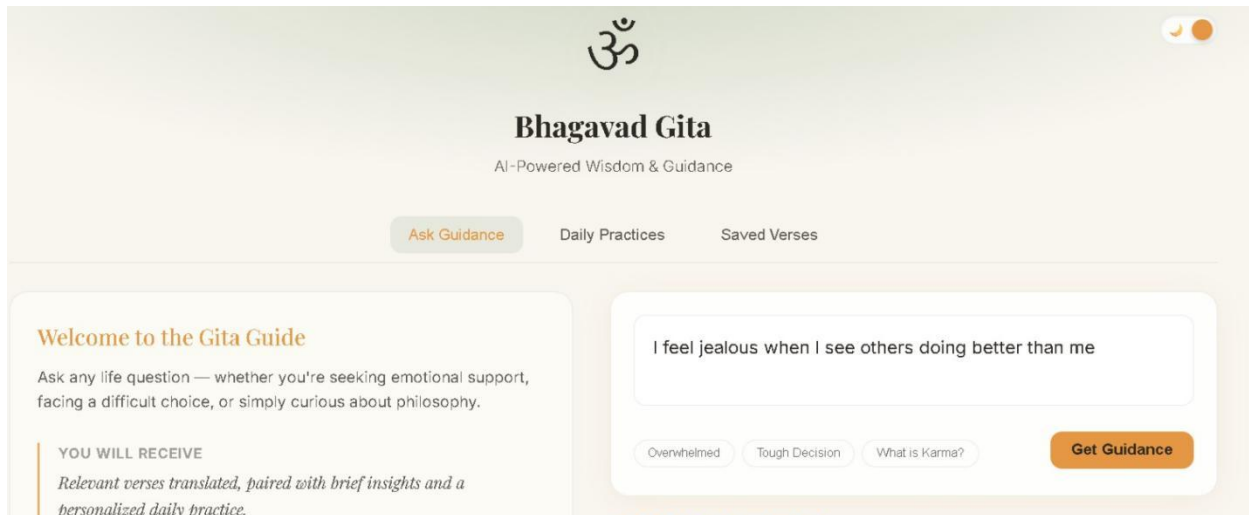


Figure 4.1 Homepage with user input and guidance request

The system's intent analysis module identifying the user's emotional state as envy and categorizing the query under life guidance. It presents a relevant Bhagavad Gita verse along with a concise interpretation and actionable advice to help the user manage the emotion.

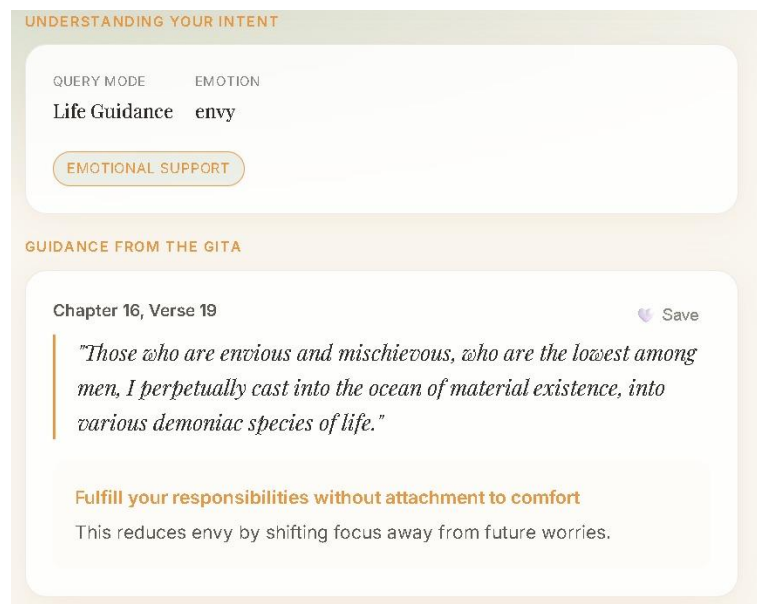


Figure 4.2 Intent analysis with detected emotion and Gita guidance

This shows the daily practice section providing structured, step-by-step reflective exercises. These actions are designed to help users manage emotional challenges and apply Bhagavad Gita teachings through practical, real-world steps.

MOVING FORWARD

"Redirect your energy inward: list one strength that is uniquely yours"

Pause & Control — Duty for personal emotional difficulty

- 1 Choose one response that serves your need without harming the situation. Wait 1 hour before executing it.
- 2 Identify your most important responsibility today. Do it first, before anything optional. Don't wait for motivation — duty comes before feeling.
- 3 Write what triggered the anger in one sentence. Then write: "What I actually need here is ____."
- 4 Pause for 3 minutes. Ask yourself: 'Am I reacting from habit or from understanding?' Write your honest answer.
- 5 Take one small real-world action today based on this reflection (e.g., write it down, make a decision, or reach out to someone).

 Share Guidance

Figure 4.3 Daily practice steps for emotional reflection and growth

Table 4.1 Performance Metrics Summary

METRIC	VALUE
FAISS index size	657 verses
Sample size (evaluation queries)	12
Top-K retrieval	9
Precision@K	0.22
Recall@K	0.45
Mean Reciprocal Rank (MRR)	0.65
Mean Average Precision@K (MAP@K)	0.33
NDCG@K	0.44

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project demonstrates the feasibility of building a Hybrid Retrieval-Augmented Generation (HybridRAG) system that maps modern life situations (e.g., stress, burnout, grief, promotion anxiety) to the Bhagavad Gita’s teachings using a combination of Vector Database semantic retrieval and a Knowledge Graph (KG). The vector layer ensures broad, natural-language accessibility across the entire scripture by retrieving relevant verses even when user queries contain contemporary phrasing. In parallel, the KG layer enforces a structured and explainable mapping from Situation → Emotion → Principle → Verse, reducing ambiguity and preventing inconsistent interpretation by grounding recommendations in a consistent ontology.

By designing a master ontology with fixed labels and relationships, the system avoids fragmentation and enables repeatable reasoning paths that can be audited and improved over time. As a result, the system moves beyond simple “verse search” and provides an interpretable framework for linking a user’s lived experience to specific spiritual principles. Overall, this work validates that HybridRAG can serve as a scalable foundation for spiritually grounded guidance systems where both semantic relevance (vector search) and doctrinal structure (knowledge graph) are required for trustworthy outputs.

5.2 Future Work

The current implementation establishes the foundational pipeline for verse retrieval and structured mapping, but several future enhancements can transform it into a more scalable, personalized, and real-world effective guidance platform. Future work includes constructing a full-scale Knowledge Graph for the entire Bhagavad Gita, ensuring that every verse is connected to relevant principles, emotions, and real-life contexts to enable deeper multi-hop reasoning. The system can also evolve from informational guidance to behavior-focused intervention by introducing nodes such as Practice, Routine, Reflection Prompt, and Barrier to generate actionable micro-practices for users. To improve reliability and scientific credibility, comprehensive evaluation frameworks using metrics like Top-k verse relevance, groundedness, explainability coverage, ontology compliance, and user-reported outcomes should be developed along with expert-validated benchmark datasets. Since users may interact with the system during emotionally vulnerable situations, future versions should incorporate safety mechanisms capable of detecting high-risk contexts and recommending professional support when necessary. Collectively, these enhancements can evolve the project from a technically correct HybridRAG prototype into a scalable, explainable, and behaviorally effective spiritual guidance system for practical daily life application.

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